FLOODRESQ

WHAT WE ARE EVALUATING

- **Ergonomics:** Is the grip secure when hands are wet or muddy?
- **Physical Effort:** Is high suction pressure required for children or the elderly?
- **Visual Feedback:** Can the user see the filter working through the tube?

CONTROLLED LAB ENVIRONMENT

- **The Setting:** A laboratory environment simulating flood conditions (dim lighting, wet surfaces).
- **Why Not Real World?:** Testing in an actual flood is unethical and dangerous.
- **Level of Control:** High Control.

1. We control the water "turbidity" (dirtiness).
2. We control the product weight.
3. We can stop the test immediately if safety is compromised.

MEASURING SUCCESS

Quantitative (The Numbers):

- Time-to-Drink: Time taken (in seconds) from picking up the device to drinking successfully.
- Error Rate: Number of attempts using the wrong end or failing to uncap.

Qualitative (The Feedback):

- User Observation: Facial expressions (struggle, confusion, relief).
- Post-Test Survey: Rating "Confidence in Safety" on a 1–5 scale.

USABILITY VS. USER EXPERIENCE

USABILITY

(Performance)

- Effectiveness: Did the water flow?
- Efficiency: Was it fast to deploy?
- Safety: Did the mouthpiece remain uncontaminated?

USER EXPERIENCE

(Feeling)

- Trust: Does the transparent design reassure the user?
- Psychological Comfort: Does the device reduce panic?

Limitations of the Evaluation

- **Safety Ethics:** We cannot use actual toxic floodwater; we must use "mock" contaminants (safe substitutes like sterile soil).
- **Resource Limits:** We are testing a single prototype; durability failure ends the test.
- **Participant Bias:** Test subjects are engineering students, not random people in public.